European Polymer Journal 241 (2025) 114384

Contents lists available at ScienceDirect

European Polymer Journal

journal homepage: www.elsevier.com/locate/europolj

Influence of end-group functionality and composition on the antifungal activity and biocompatibility of polyacrylamide-based polymers

Hatu Gmedhin [a,b,c], Peifeng Wu [c], Zi Gu [a,c], Megan D. Lenardon [b,*], Cyrille Boyer [a,c,**]

[a] Cluster for Advanced Macromolecular Design (CAMD) and Australian Centre for NanoMedicine (ACN), School of Chemical Engineering, UNSW, Sydney, NSW 2052, Australia
[b] School of Biotechnology and Biomolecular Sciences, UNSW, Sydney, NSW, 2052, Australia
[c] School of Chemical Engineering, UNSW, Sydney, NSW 2052, Australia

ARTICLE INFO

Keywords:
Antimicrobial polymers
RAFT polymerization
RAFT end-group
Antifungal activity

ABSTRACT

Drug-resistant invasive fungal infections are a growing health threat, prompting the development of synthetic antifungal polymers as alternative therapeutic platforms. Here, we report the systematic design of polyacrylamide terpolymers synthesized via photoinduced electron/energy transfer–reversible addition-fragmentation chain transfer (PET-RAFT) polymerization. Molecular weight, varied by controlling chain length, determined antifungal selectivity: shorter polymers exhibited superior selectivity against *Candida* species. Chain-length thresholds were identified for *Candida albicans* and *Candida tropicalis*, while *Candida parapsilosis* responded most strongly to the lowest-molecular-weight (MW) analogs. To investigate the combined influence of the polymer end-group functionality and molecular composition, three distinct chain-transfer agents (RAFT agents) were employed, and composition was systematically tuned. Selective ω–end-group modification was then performed to introduce either hydroxyl (–OH) or hydrophobic heptyl groups. End-group functionality, together with composition control, governed structure–activity relationships, enabling the precise tuning of antifungal activity and biocompatibility. This integrated design strategy establishes molecular-level principles to guide the rational development of antifungal polymers.

1. Introduction

Invasive fungal infections are increasingly recognized as a global health crisis, now affecting over 6.5 million individuals and leading to more than 3.7 million deaths worldwide annually. [1–3] Among these, bloodstream infections caused by *Candida* species rank among the top ten causes of nosocomial pathogenic infections in intensive care units, with a mortality rate exceeding 35 %. [4,5] While *Candida albicans* remains the dominant species, non-*albicans Candida* species, such as *Candida tropicalis* and *Candida parapsilosis*, as well as multidrug-resistant *Candida auris* (recently reclassified under the nomenclature *Candidozyma auris*) [6], have become increasingly prevalent. [7,8] Current antifungal drugs for invasive fungal infections are limited to three classes, constrained by host toxicity and rapidly evolving drug resistance. [9–13].

Synthetic mimics of antifungal peptides have been explored as promising alternatives to conventional antifungal agents. [14–16] Inspired by host defence peptides, these materials mimic cationic and amphiphilic features, enabling selective disruption of fungal pathogens while minimizing off-target toxicity. [17,18] Advances in living/controlled polymerization techniques, particularly reversible addition-fragmentation chain transfer (RAFT) polymerizations, provide access to such polymers with precise control over molecular weight, a narrow dispersity, and preserve end-group fidelity. [19–23] RAFT polymerization also leaves a chemical imprint at both termini: the α–ends of the polymer correspond to the R-group of the chain transfer agents (CTAs), and the ω–ends correspond to the Z-group of the CTAs. [24,25] Notably, synthetic antifungal polymers often consist of short polymer chains, making their bioactivity highly sensitive to variation in α– and ω–end terminal groups. [26–28].

The structure–activity relationships (SARs) in antifungal polymers are influenced by several factors, including molecular weight, [29] the

* Corresponding author.
** Corresponding author at: Cluster for Advanced Macromolecular Design (CAMD) and Australian Centre for NanoMedicine (ACN), School of Chemical Engineering, UNSW, Sydney, NSW 2052, Australia.
E-mail addresses: m.lenardon@unsw.edu.au (M.D. Lenardon), cboyer@unsw.edu.au (C. Boyer).

https://doi.org/10.1016/j.eurpolymj.2025.114384
Received 21 September 2025; Received in revised form 28 October 2025; Accepted 1 November 2025
Available online 6 November 2025
0014-3057/© 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

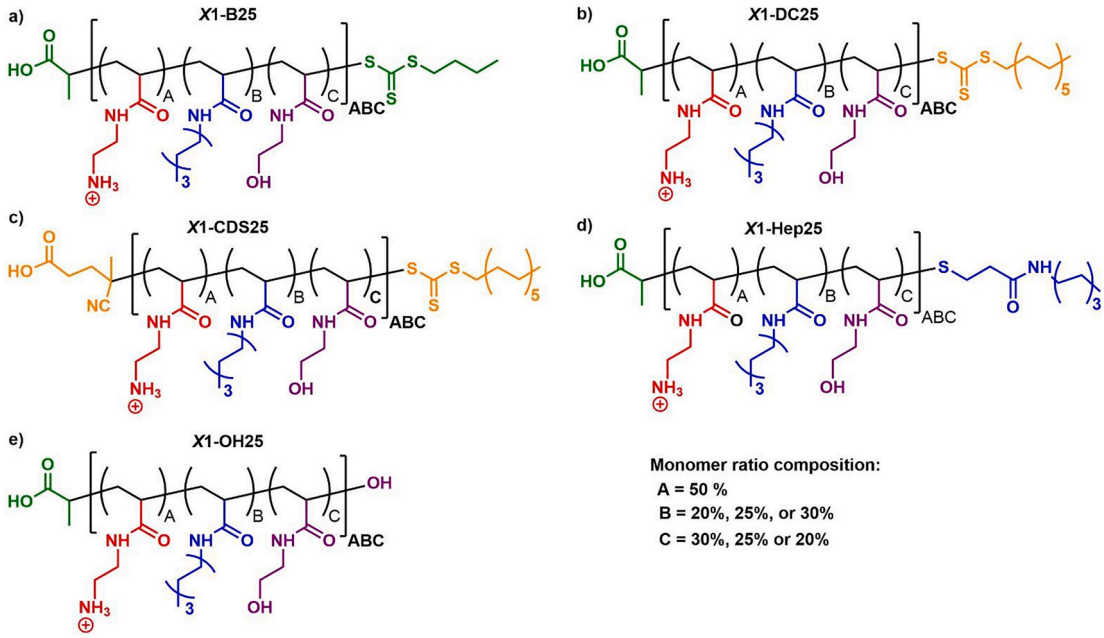


Fig. 1. Chemical structures of the antifungal ternary polyacrylamide polymers prepared in this study. Each polymer consists of three monomers: cationic *N*-(2-aminoethyl) acrylamide (AEm, shown in red), hydrophobic group *N*-heptyl acrylamide (HepAm, blue), and hydrophilic *N*-hydroxyethyl acrylamide (HEAm, purple). The polymer code ***X*1-B25** is defined as follows: *X*1 indicates the fold-number targeted degree of polymerization (e.g., *X*1 corresponds to $X_n \approx 20$), B represents the *ω*–end terminal derived from RAFT agents and its modification (e.g., *B* refers to 2-(butylthiocarbonothioylthio) propanoic acid, BTPA); and 25 denotes the 25 mol% fraction of the hydrophobic monomer (HepAm) incorporated in the terpolymers. a) *X*1-B25 represents a terpolymer synthesized with one-fold X_n using BTPA as RAFT agent and 25 % HepAm content. **b)** *X*1-CD25 is a terpolymer synthesized using 2-(dodecylthiocarbonothioylthio) propionic acid (DCTPA) as RAFT agent ("DC"), **d)** *X*1-DCS25 polymer is prepared using (4-cyano-4-((dodecylthio)carbonothioyl)thio)pentanoic acid CDSTPA as RAFT agent ("CDS"), **e)** *X*1-Hep25, the RAFT *Z*–end group is modified to introduce HepAm monomers ('Hep') at the *ω*–end of the polymer **f)** *X*1-OH25 refers to a terpolymer in which the ω–end groups are modified with a hydroxyl ("OH") terminal group.

nature of cationic groups, [30–32] global amphiphilic balance, [33] and topology. [34,35] End-groups have also been shown to modulate activity, complementing these established factors. [36] For example, *α*–end-groups of polymethacrylate copolymers have been reported to modulate hemolytic activity, while *ω*–end-groups play a role in antifungal activity. [37] Related studies have further demonstrated that variation in block arrangement related to the *α*–and *ω*–end-groups, as well as the hydrophobicity, affects both hemocompatibility and antifungal activity, respectively. [34,38].

Comparative insights can be drawn from antifungal peptides, where their terminal modifications exert a marked influence on antifungal activity. [39,40] Strategic terminal modification, including N-terminal glycine capping and C-terminal amidation or hydrophobic tagging, has been shown to enhance antifungal potency and therapeutic efficacy.

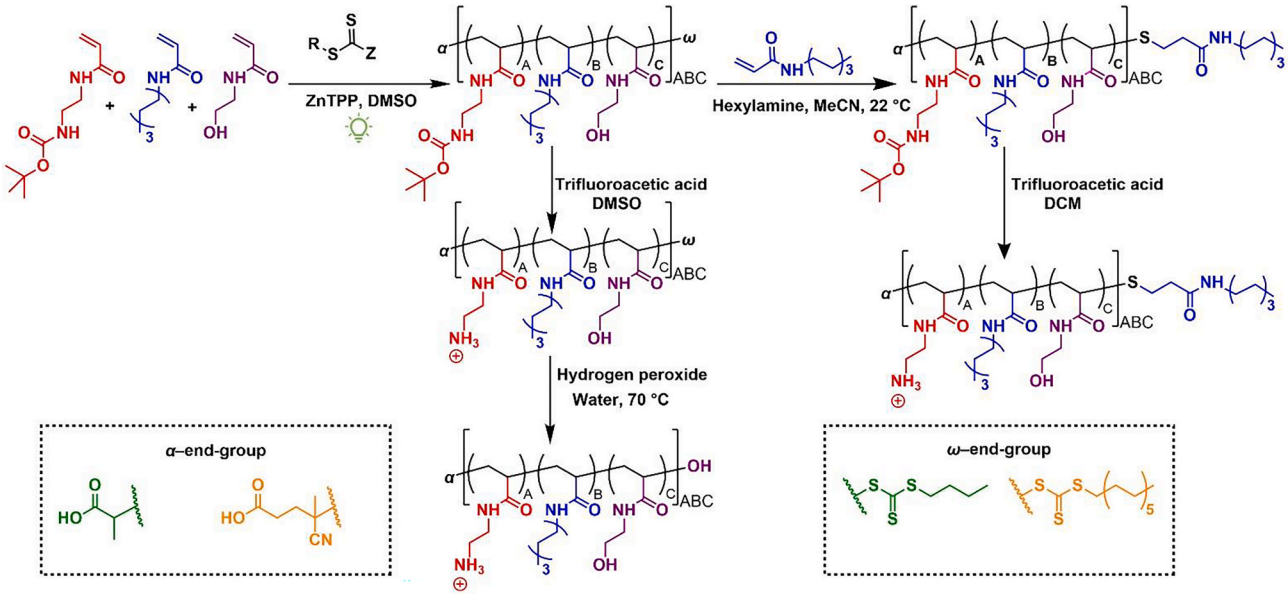
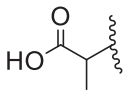
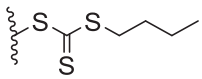
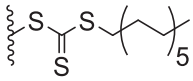
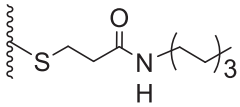
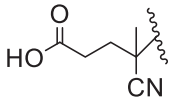
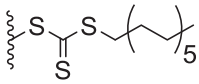


Fig. 2. Synthesis of antifungal terpolymers via PET-RAFT polymerization using Boc-AEm, HepAm, and HEAm monomers. Terpolymers were synthesized with varied α-end groups and ω–end groups and further modified at the ω–end to introduce hydrophilic or hydrophobic terminal functionalities.

Table 1
Characterization of terpolymers synthesized in this study.

End group α	*ω*	*ω*–ClogP	Polymer	*Molar Composition [a] A: B: C*	X_n $_{NMR}$ [b]	M_n [c] (g mol^{-1})	M_n [d] (g mol^{-1})	Đ [e]	D_h [f] (nm)	ζ [f] (mV)	T%[g]
		3.8	*X*-B25	49:26:24	10	2000	3500	1.14	4.4	41.0	91.2
		3.8	*X*1-B25	50:25:25	20	3800	5600	1.11	1.9	60.2	94.7
		3.8	*X*2-B25	50:25:25	39	6900	12,000	1.08	1.4	33.2	96.2
		3.8	*X*3-B25	52:24:24	80	12,100	22,500	1.08	7.5	44.0	95.9
		3.8	*X*4-B25	50:26:24	161	22,500	39,500	1.10	10.8	30.1	96.4
		3.8	*X*1B-20	50:20:30	20	3700	5700	1.13	28.2	7.8	96.1
		3.8	*X*1B-30	48:30:22	20	3800	6000	1.13	1.2	37.1	95.7
		−0.8	*X*1-OH25	50:25:25	20	3800	n/a	n/a	20.5	38.2	96.3
		−0.8	*X*1-OH30	48:30:22	20	3800	n/a	n/a	30.0	60.7	94.0
		8.0	*X*1-DC20	51:20:29	20	3850	6300	1.13	6.3	15.7	96.0
		8.0	*X*1-DC25	50:25:25	20	3900	6500	1.12	2.9	32.5	95.1
		8.0	*X*1-DC30	51:29:21	20	3900	6100	1.1	4.3	57.3	93.0
		2.8	*X*1Hep20	51:20:30	20	3850	6000	1.13	35.1	10.8	96.2
		2.8	*X*1Hep25	50:25:25	20	4000	6200	1.12	26.5	37.2	95.7
		8.0	*X*1-CDS25	49:25:26	20	3900	6600	1.10	29.3	52.2	96.1
		8.0	*X*1-CDS30	52:27:21	20	4000	6800	1.10	77.3	62.7	96.1

Note: Chemical structures and *ω*–clogP (calculated partition coefficient values, rounded to one decimal place) were obtained using ChemDraw, version 23.1.2. **a)** Composition determined by ^{1}H NMR spectroscopy (**Supporting Information, Figs. S4 and S5**); **b)** The degree of polymerization (X_n) from ^{1}H NMR spectra (**Supporting Information, Figs. S9 and S10**); **c)** M_n of the Boc-protected polymers was obtained via ^{1}H NMR spectroscopy (**Supporting Information, Fig. S5**); **d)** The M_n values were also determined by SEC using *N, N'*-dimethylacetamide eluent, PMMA standards (**Supporting Information, Fig. S15**); **e)** Dispersity (*Đ*) was derived from SEC data; **f)** The hydrodynamic diameter (D_h) and zeta potential (*ζ*) of Boc-deprotected were measured by dynamic light scattering; **g)** T% refers to turbidity measurements recorded as percentage transmittance at 600 nm.

[41,42] Similar trends are evident in synthetic polymers, where nylon-3 polymers exhibit altered antifungal activity and hemocompatibility depending on terminal groups. [43] Further, post-polymerization modification of polymer *ω*–end-group in amphiphilic methacrylate copolymers to cyanoisobutyl or aminoethyl cyanopentanoate markedly diminished both antibacterial potency and hemolytic activity. [44] Collectively, these findings highlight terminal chemistry as a versatile platform for probing SARs and implementing post-polymerization strategies. [44,45] Despite these advances, systemic approaches to investigate RAFT end-group modifications, while simultaneously controlling molecular composition, remain largely underexplored.

In this study, we present a library of well-defined polyacrylamide terpolymers synthesized via photo-induced electron/energy transfer-RAFT (PET-RAFT) polymerization. We evaluated the interplay between *α*– and *ω*–end-group polymer and chain hydrophobic/hydrophilic composition on antifungal efficacy and biocompatibility. In this work, the *α*– and *ω*–end-groups were used for the termini of the polymer chain, whereas the terms, R- and Z-groups, were explicitly employed when discussing the RAFT agents. By comparing polymers synthesized using three distinct RAFT agents and selectively modifying the *ω*–end-groups with hydrophilic or hydrophobic functionalities (Fig. 1), we uncovered a SAR that governs therapeutic performance. This highlights the crucial role of end-group chemistry as a tunable design parameter. Our findings demonstrate that the development of next-generation antifungal polymers with high selectivity requires optimizing both the terminal and polymer compositions.

2. Results and discussion

2.1. Synthesis and characterization of ternary polyacrylamides

A library of well-defined ternary amphiphilic polymers was generated via PET-RAFT polymerization using 2-(butylthiocarbonothioylthio) propanoic acid (BTPA) as the chain transfer agent and 5,10,15,20-tetraphenyl-21H,23H-porphine zinc as photocatalyst, under green light irradiation conditions (λ = 530 nm, I_0 = 1.75 mW cm^{-2}) (Fig. 2). Monomer selection was guided by prior optimization studies on antifungal polyacrylamides to balance antifungal activity and biocompatibility (**Supporting information, Table S1**). The cationic precursor, *tert*-butyl (2-acrylamidoethyl) carbamate (Boc-AEm), can be deprotected in a single step to afford *N*-(2-aminoethyl) acrylamide (AEm), which is predominantly protonated at physiological pH. Comparative studies indicate primary ammonium ($-N^+H_3$) provides superior selectivity relative to guanidinium- and imidazolium-based analogues. [46] The cationic monomer content was fixed at 50 mol %, based on previous studies in which systematic variation of A (40/50/60 mol %) identified this fraction as optimal. [14,28,46] The hydrophobic monomer *N*-heptyl acrylamide (HepAm) introduces linear seven-carbon (C_7) chain, previously identified as an efficient hydrophobic group to provides optimal compromise between potency and biocompatibility. [14,28] The neutral hydrophilic *N*-hydroxyethyl acrylamide (HEAm) limits polymer aggregation in water and media and reduces hemolysis without compromising antifungal activity. [14,47] Boc-Aem and HepAm were synthesized via carbodiimide-mediated amine coupling, as previously established, whereas HEAm was used as received (**Supporting Information, Figs. S1–S3**). [38].

Ternary polyacrylamides copolymer with targeted number average degrees of polymerization ($X_n \approx$ 10, 20, 40, 80, and 160, denoted as *X*-B25, *X*1-B25, *X*2-B25, *X*3-B25, and *X*4-B25, respectively) were prepared by varying the amount of BTPA as the RAFT agent (Table 1). Size exclusion chromatography (SEC) analysis, using DMAc as the eluent and polymethyl methacrylate (PMMA) as the standard, revealed a narrow molecular weight distribution (MWD) with good agreement between the experimental and theoretical values (Table 1). However, some divergences were observed between theoretical (e.g., X2-B25, M_n = 3800 g mol^{-1}) and experimental M_n values (X2-B25, M_n = 5600 g mol^{-1}),

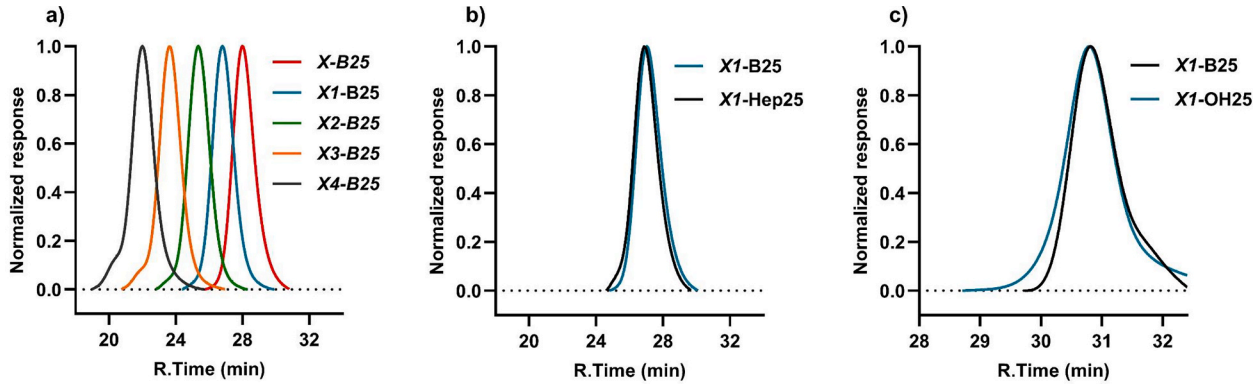


Fig. 3. Characterization of terpolymers prepared using BTPA as RAFT agent before and after Z-end-group modifications. **a)** Normalized SEC traces of BTPA-terminated polymers (*X*–B25 series) in DMAc with PMMA calibration standards, demonstrating increasing X_n from *X* to 25 to X25. **b)** SEC traces of *X*1-B25 before (blue line) and *X*1-Hep25 after aminolysis/thiol-ene (block line), showing no change in retention time upon BTPA-to-heptyl end-group substitution. **c)** Aqueous SEC traces of *X*1-B25 before (blue line) and after hydroxyl-modified *X*1-OH25 (block line) (1 % TFA, 1 M NaCl with 2-vinylpyridine calibration standard). The slight leftward shift in *X*1-OH25 indicates increased hydrophilicity and hydrodynamic volume.

which were attributed to the difference in hydrodynamic volume between the amphiphilic terpolymers and the PMMA standard, consistent with previous reports (Table 1). [38,46] Overall, the polymer displayed a narrow molecular weight distribution with dispersity (*Đ*) values close to 1.10, although a slightly broader distribution was observed for shorter chain length (*X*-B25, *Đ* = 1.14) (Table 1). ^{1}H NMR spectroscopy confirmed the successful incorporation of all three monomers, with calculated compositions closely matching the theoretical feed ratios calculated both before and after polymerization (**Supporting Information, SI, Figs. S4 and S5**).

2.2. RAFT end-group variation in terpolymer synthesis

To examine the impact of end-group variation on polymer properties, we synthesized additional polymers with a target degree of polymerization *(X_n)* of approximately 20. These polymers had compositions similar to those made with BTPA, but were prepared using two additional RAFT agents, 2-(dodecylthiocarbonothioylthio propionic acid (DCTPA) and 4-cyano-4-(dodecylthio)carbonothioyl)thio)pentanoic acid (CDSTPA). Unlike BTPA, these RAFT agents incorporate a dodecyltrithiocarbonate as the *ω*–end-groups of the polymer chain. In addition, CDSTPA incorporates a cyanovaleric *α*–end, in contrast to a carboxylic acid end-group (**Supporting Information, Figs. S6–S8**). Under green PET-RAFT(λ = 530 nm, I_0 = 1.75 mW cm^{-2}), monomer conversion reached > 98 % within 16 h for DCTPA and within 24 h for CDSTPA (**Supporting Information, Fig. S9**). In a previous report, [38] we demonstrated that the three monomers are statistically distributed along the polymer chain, with minimal composition drift under these conditions. The final polymer compositions and X_n values aligned with our targets (Table 1, **Supporting Information, Fig. S4, S5, and S10**). SEC confirmed the successful synthesis of polymers with narrow dispersity (*Đ* < 1.13) and consistent M_n values (5.7 – 6.8 kg mol^{-1}). Subsequently, post-polymerization deprotection of the Boc group using trifluoroacetic acid (TFA) was confirmed by the complete disappearance of the *tert*-butyl proton signal at δ = 1.4 ppm in the ^{1}H NMR spectrum (**Supporting Information, Fig. S11**).

2.3. Post-modification of RAFT end-groups

To complete our investigation, we selectively removed the polymer *ω*–end-groups from polymers prepared with BTPA. We achieved this using two methods: (i) by *in-situ* aminolysis and thiol-ene reactions, or (ii) by oxidation with hydrogen peroxide (Fig. 3). *X*1-B25 was selected as a model polymer for these reactions. For the aminolysis and thiol-ene reaction, *X*1-B25′s RAFT end-group was cleaved in the presence of hexylamine and *N*-heptyl acrylamide. SEC analysis of *X*1-B25 before and after this reaction (Fig. 3b) showed no significant change in molecular weight or dispersity, confirming that the end-group cleavage and subsequent modification step did not compromise the polymer backbone or result in the formation of disulfide. Further validation of end-group removal was provided by UV detection at 310 nm via SEC, which revealed the absence of the characteristic signal in the functionalized polymer (*X*1-Hep25) compared to the unmodified *X*1-B25 (**Supporting Information, Fig. S12c**). To further validate the success of polymer *ω*–end-groups modification, ^{1}H NMR spectroscopy was employed to analyze terpolymers before and after RAFT modification (**Supporting Information, Figs. S12a and S13**). The unmodified terpolymer *X*1-B25 displayed the RAFT agent signals, such as CH in the adjacent trithiocarbonate and CH_3 of the R group. After aminolysis of trithiocarbonate and thiol-ene with HepAm, the presence of new signals around 2.5–2.7 ppm emerged, confirming the successful thiol-ene reaction (**Supporting Information, Fig. S12a-c**).

To explore an alternative *ω*–end group strategy, we also followed an established protocol, treating *X*1-B25 with hydrogen peroxide at 70 °C to replace the trithiocarbonate end group with an –OH group. [48] UV–Vis spectroscopy (**Supporting Information, Fig. S12f**) confirmed the successful elimination of trithiocarbonate by the absence of signal at 310 nm. Further hydroxyl-functionalization analogues revealed an –OH signal at 3.17 ppm, indicating successful incorporation of hydrophilic –OH terminus (**Supporting Information, Fig. S12c-f and S14**). SEC performed in aqueous media (1 % TFA, 1 M NaCl) revealed a slight leftward shift in retention time for X1-OH25 compared to X1-B25, while maintaining the same molecular weight and dispersity (Fig. 3). This shift indicates a modest increase in hydrodynamic volume upon RAFT end-group modification, consistent with the replacement of a hydrophobic BTPA group with a more hydrophilic hydroxyl functionality. Overall, these findings confirm the successful post-modification of RAFT end-groups, indicating that the one-pot terminus transformation was achieved efficiently without compromising polymer integrity or introducing undesired coupling reactions. [49].

2.4. Aqueous media characterization

All synthesized polymers exhibited good aqueous solubility (1 mg mL^{-1}) in Milli-Q water, forming clear, non-turbid solutions (Table 1). UV–vis spectroscopy of *X*-25 copolymers showed a decrease in trithiocarbonate peak (C=S bond at 310 nm) in solution stored in Milli-Q water over 24 h (**Supporting Information, Fig. S16**), consistent with potential end group degradation in the presence of primary amine. [49–51] However, the trithiocarbonate absorbance remained stable in RPMI-

Table 2
Values of MIC of the increasing molecular weight polymers against *Candida* species.

Polymers or Antifungal drugs	Composition [a] A: B: C	X_n^b NMR	MIC$_{90}$ (μg mL^{-1}) [c] *ca1*	*ca2*	*ct*	*cp*
X-B25	49:26:24	10	32	16	32	64
*X*1-B25	50:25:25	20	16	8	16	128
*X*2-B25	50:25:25	39	256	128	256	>512
*X*3-B25	52:24:24	80	>512	512	>512	>512
*X*4-B25	50:26:24	161	>512	>512	>512	>512
Fluconazole			0.25	0.25	>4	2–4
Amphotericin B			0.5–1	0.5–1	1	1
Caspofungin			0.5	0.25	0.25	0.5

Note: a) Polymer composition (mol%) of A = *N*-(2-aminoethyl) acrylamide (AEm), B = heptyl acrylamide (HepAm), and C = hydroxyethyl acrylamide (HEAm). b) X_n denotes the degree of polymerization determined by ^{1}H NMR. c) MIC$_{90}$: refers to the minimum concentration required to inhibit greater than 90 % of fungal growth. Fungal strains tested include *C. albicans* (*ca*1 = SC5314, *ca*2 = s20122.073), *C. tropicalis* (*ct* = SCS501148), and *C. parapsilosis* (*cp* = ATCC22019).

1640 medium (pH 4), which was used for antifungal susceptibility testing (**Supporting Information, Fig. S17**). Dynamic light scattering (DLS) analysis of polymer solutions (1.5 mg mL^{-1} in Milli-Q water) revealed that the hydrodynamic diameter (D$_h$) varied with both hydrophobic content and end group functionality (Table 1). The BTPA-derived polymer with 25 % hydrophobic monomer content showed a D$_h$ of 1.9 nm, while adjustments to 20 % and 30 % resulted in D$_h$ values of 28.2 nm and 1.2 nm, respectively (Table 1). Polymer modified with terminal hydroxyl groups exhibited a larger diameter (20.5 – 30.0 nm), whereas those bearing carboxyl termini (DCTPA series) ranged from 2.9 to 6.3 nm (Table 1). Incorporation of heptyl acrylamide units increases D$_h$ to 26.5 – 35.1 nm. The largest aggregations were observed for polymers synthesized using CDSTPA as the RAFT agent, reaching 77.3 nm at 30 % hydrophobicity content (Table 1). These results demonstrate that both backbone hydrophobicity and end-group properties influence the polymer's hydrodynamic size in aqueous solution. Further, zeta potential (ζ) measurements confirmed that all deprotected terpolymers possessed a net positive surface charge, validating successful removal of Boc groups and the formation of primary ammonium functionalities. This positive surface charge potential is expected to enhance electrostatic interactions with the fungal membrane cell surface, a key contributor to antifungal activity. [52].

2.5. Molecular weight influence on antifungal activity

The antifungal activity of the synthesized polymers was evaluated by determining the minimum inhibitory concentrations (MICs) against three *Candida* species, which cause invasive *Candida* infections. [53–55] Since *Candida albicans* strains can be genetically and phenotypically diverse [56], we tested the *C. albicans* reference strain (SC5314) [57] as well as a *C. albicans* clinical isolate (s20122.07) [58]. A clinical isolate of *C. tropicalis* SCS501148) [59] and a type strain of *C. parapsilosis* (ATCC22019) [60] were tested to explore the potential extension of end-group–dependent activity trends to *Candida* species beyond *C. albicans* (**See Supporting Information, Table S2**). The MICs were determined according to Clinical and Laboratory Standards Institute (CLSI) guidelines with minor modifications. [61] MIC was defined as the lowest concentration of drug that inhibited greater than 90 % fungal growth after 24 h of incubation. The standard antifungal agents caspofungin, fluconazole, and amphotericin B were employed as controls (Table 2) and exhibited expected susceptibility profiles. [14,28,38].

To establish an operational window before investigating the end-group effect, we isolated the molecular weight (MW) as the sole variable, given that polymer end-group contribution scales approximately as $\sim 2/X_n$. A composition-matched series was synthesized using BTPA RAFT agent across a range of $X_n \approx$ 10–160 (Table 2) to evaluate the relationship between MW and antifungal activity. Against both *C. albicans* strains, potency peaked near $X_n \approx 20$ (X1-B25), consistent with our previous report. [14] The lower-MW analogue (*X*-B25; $X_n \approx 10$) exhibited reduced activity, while higher-MW variants (Xn ≈ 40–160; Table 2) showed a marked decline in antifungal potency. Similar chain length thresholds have been reported for polyamide polymers and poly(trimethylenimine) against *C. albicans.* [15,29].

Table 3
Antifungal activity of polymers with different *α*– and *ω*–end-groups and hydrophobicity against *Candida* species.

Polymers	Composition A: B: C	MIC$_{90}$% (μg mL^{-1}) [c] *ca1*	*ca2*	*ct*	*cp*	*ω*–end groups
*X*1-B20	50:20:30	64	32	32	256	C$_4$
*X*1-B25	50:25:25	16	8	16	128	C$_4$
*X*1-B30	48:30:22	8–16	4–8	16	64	C$_4$
*X*1-OH25	50:25:25	64	32	64	256	–OH
*X*1-OH30	48:30:22	16	8	32	128	–OH
*X*1-Hep20	50:20:30	64	64	64	512	HepAm
*X*1-Hep25	50:25:25	32	16	32	128	HepAm
*X*1-DC20	51:20:29	16	8	16–32	128	C$_{12}$
*X*1-DC25	50:25:25	32	8	32	64	C$_{12}$
*X*1-DC30	51:29:21	32	4–8	16–32	128	C$_{12}$
*X*1-CDS25	49:25:26	32–64	16	32	128	C$_{12}$
*X*1-CDS30	52:27:21	16	8	16	32	C$_{12}$

Note: a) Polymer composition (mol%) of A = *N*-(2-aminoethyl) acrylamide (AEm), B = heptyl acrylamide (HepAm), and C = hydroxyethyl acrylamide (HEAm). b) X_n denotes the degree of polymerization determined by ^{1}H NMR spectroscopy. c) MIC$_{90}$: refers to the minimum concentration required to inhibit greater than 90 % of fungal growth. The *ω*–end corresponds to butyl (C$_4$) or dodecyl (C$_{12}$) alkyl groups, post-modified with hydroxyl (–OH) or HepAm. Fungal strains tested include *C. albicans* (*ca*1 = SC5314, *ca*2 = s20122.073), *C. tropicalis* (*ct* = SCS501148), and *C. parapsilosis* (*cp* = ATCC22019).

In contrast, *C. parapsilosis* displayed the lowest MIC (64 μg mL^{-1}) for *X*-B25 ($X_n \approx 10$), while the highest MW variants ($X_n \geq 40$) were inactive (*X*2-B25, MIC > 512 μg mL^{-1}: Table 2). Notably, MW–activity relationships are polymer type- and fungal species-dependent, which is supported by previous results in the literature. For example, chitosan-based cationic polymers (75.7, 140.5, and 247.8 kg mol^{-1}), showed that higher MW correlated with improved antifungal activity against strains of *Candida* species (e.g., *C. albicans, C. tropicalis, and C. parapsilosi*). [62] In contrast, in cationic ionenes, it has been recently reported that lower MW (2.8 and 4.4 kg mol^{-1}) yielded higher potency against *C. albicans* and *C. tropicalis* than their higher MW analogues (9.1 and 30.6 kg mol^{-1}). [63] Accordingly, we selected $X_n \approx 20$ (X1-B25) as the baseline for end-group comparisons, based on the activity trends for *C. albicans* and *C. tropicalis* (Table 2).

2.6. The influence of RAFT α–end groups on antifungal activity

To explore structural-activity relationships beyond MW, terpolymers with varied RAFT end groups were evaluated. The polymers tested (Table 3) were synthesized: with a fixed degree of polymerization ($X_n \approx 20$ – polymer name starting with *X*1); different RAFT agents to alter the polymer chain end-groups (BPTA – B in polymer name, DCTPA – CD in polymer name, CDSTPA – CDS in polymer name); varied polymer composition to adjust amphiphilicity (ratio of cationic: hydrophobic: hydrophilic monomers indicated by A:B: C; mol% fraction of hydrophobic monomers shown by the number (20, 25 or 30) in the polymer name); and, selective modification the RAFT end groups (Z-end-group modified to an HepAm monomer – Hep in polymer name, Z-end-group modified with an hydroxyl terminal group –OH in the polymer name).

Terpolymers synthesized with the three distinct RAFT agents (B, DC, and CDS as indicated in the polymer name) exhibited potent antifungal activity across all tested *Candida* species, with MICs ranging from 4 to 8 to 64 μg mL^{-1} (Table 3). To specifically assess the influence of *α*–end-groups, we compared polymers bearing a carboxylic acid (–COOH, *X*1-

DC25 and *X*1-DC30) with those bearing a cyanovaleric acid (CVA) group (*X*1-CDS25 and *X*1-CDS30). We observed subtle differences in the polymer with a carboxylic acid *α*–end-group (e.g., X1-DC25, MIC = 32 μg mL-^{1}) against *C. albicans* SC5314 compared to the corresponding CVA group-terminated polymer (*X*1-CDS25, MIC = 32–64 μg mL^{-1}) (Table 1). A similar observation has been reported previously on binary primary amine-based polymethacrylate, whereas more pronounced effects were detected in guanidinium-based analogs. [37] Taken together, these findings indicate that the limited influence of *α*–end-group identity reported in binary polymethacrylates is also maintained in our polyacrylamide terpolymer variants, with overall antifungal activity remaining largely unaffected.

2.7. Hydrophobicity balance and ω–end-group tuning

The infleunce of ω–end group hydrophobicity on antifungal activity was further assessed by comparing terpolymers bearing short-chain (C_4) RAFT Z–end groups (e.g., *X*1-B20/25/30) with those possessing long-chain (C_{12}) alkyl groups (*X*1-DC20/25/30). At 25 % hydrophobic content, both C4 (*X*1-B25) and C12 (*X*1-DC25) terminated polymers exhibited strong activity against *C. albicans* and *C. tropicalis*, with only a slight increase in MIC observed for *X*1-B25 against *C. parapsilosis* (Table 3). At 20 % hydrophobic content, a divergence became apparent: the C_{12}-terminated polymer (*X*1-DC20) maintained high potency (e.g., MIC = 16 μg mL^{-1} against *C. albicans* SC5314), whereas the C_4-terminated polymer (*X*1-B20) showed a four-fold reduction in activity (e.g., MIC = 64 μg mL^{-1} against *C. albicans* SC5314). Increasing hydrophobicity by 10 % produced different outcomes depending on the ω–end group. For C_4-terminated polymers, the shift from 20 % to 30 % hydrophobic content markedly enhanced activity (e.g., *X*1-B20 MIC = 64 μg mL^{-1} against *C. albicans* SC5314, compared to *X*1-DC30 MIC = 8–16 μg mL^{-1}). By contrast, C_{12}-terminated polymers showed little improvement over the same range (*X*1-DC20 MIC = 16 μg mL^{-1} against *C. albicans* SC5314, compared to *X*1-DC30 MIC = 32 μg mL^{-1}). As a study limitation, given the concordant MIC values across three repeated assays and the two-fold dilution resolution inherent to the method, composition–activity differences within this narrow hydrophobic feed range (20–30 %) are best interpreted as indicative of a local trend, while recognizing that small but reproducible shifts in polymer composition and amphiphilicity may still influence antifungal activity.

Overall, these findings suggest that the impact of polymer amphiphilicity is modulated by the ω–end group: optimal antifungal efficacy arises from a balance between backbone hydrophobicity and terminal group design, underscoring the importance of co-optimizing both parameters in antifungal polymer design.

Further *ω*–end-group modifications with hydrophilic hydroxyl (–OH) or hydrophobic HepAm units also influenced antifungal activity (Table 3). The hydroxyl-functionalized polymer displayed reduced activity compared to the non-functionalized polymer against all fungal species tested (e.g., *X*1-OH25 MIC = 64 μg mL^{-1} against *C. albicans* SC5314, compared to *X*1-B25 MIC = 16 μg mL^{-1}). Notably, restoring the hydrophobic monomer fraction to 30 % recovered activity equivalent to the non-functionalized polymer (e.g., *X*1-OH30 MIC = 16 μg mL^{-1} against *C. albicans* SC5314). The polymer with a hydrophobic terminal HepAm group (*X*1-Hep25) also exhibited a slight reduction in antifungal activity against *C. albicans* SC5314 compared to its unmodified precursor (*X*1-Hep25, MIC = 32 μg mL^{-1} vs MIC = 16 μg mL^{-1} for *X*1-B25). This reduction became more pronounced when the polymer hydrophobic content was lowered from 25 % to 20 % (X1-Hep20 MIC = 64 μg mL^{-1}). This interpretation is consistent with prior observations: reducing hydrophobicity in C_4-terminated polymers (*ω*–end-group clogP = 3.8; Table 1, **Supporting Information, Fig. S18**) resulted in a loss in antifungal activity, whereas for C_{12}-terminated (*ω*–end-group clogP = 8; Table 1), even a 10 % increase in hydrophobic monomer content produced no measurable difference in antifungal activity. Together, these results underscore an interdependence between terminal group hydrophobicity and the polymer amphiphilic balance.

Table 4
Antifungal activities (MIC, MFC), hemolytic activity, and cytotoxicity of synthesized terpolymers against *C. albicans* laboratory strain (SC5314), red blood cells (fresh defibrinated sheep blood), and fibroblast cell lines (3 T3 cells).

Polymers or Antifungal drugs	MIC_{90} (μg mL^{-1})	MFC (μg mL^{-1})	HC_{50} (μg mL^{-1})	SI_H	IC_{50} (μg mL^{-1})	SI_C
X-B25	32	128	1000	>63	256	8
*X*1-B25	16	128	>2000	>125	128	8
*X*2-B25	256	>512	>2000	0.8	<64	n/a
*X*3-B25	>512	>512	>2000	4	<64	n/a
X4-B25	>512	>512	>2000	n/a	<64	n/a
X1-B20	64	256	>2000	>63	256	4
X1-B30	8–16	64	1000	125–250	64	4
X1-OH25	64	256	>2000	>31	256	4
X1-OH30	16	256	>2000	>125	128	8
X1-Hep20	64	256	>2000	>31	256	4
X1-Hep25	32	128	>2000	>125	256	8
X1-DC20	16	128	>2000	>250	256	16
X1-DC25	32	64	>2000	>250	128	4
X1-DC30	32	64	>2000	>250	64	2
X1-CDS25	32–64	128	>2000	>250	128	2
X1-CDS30	16	64	500	7.8	128	8
Fluconazole	0.25	>4	>125	>500	64	256
Amphotericin B	0.5–1	2	16	16	<64	n/a
Caspofungin	0.25	>2	>125	>250	n/a	n/a

Note: MIC_{90} represents the minimum concentration required to inhibit at least 90 % growth of a *C. albicans* laboratory strain (SC5314) The Minimum Fungicidal Concentrations (MFC), determined after 48 h of incubation at 30 °C, are defined as the lowest concentration of an antifungal agent required to kill a particular fungus, assessed as no growth after plating and incubating a sample taken from the wells of the MIC assay plates for 48 h at 30 °C. HC_{50} was assessed at a maximum polymer concentration of 2000 μg mL^{-1} and indicates the concentration at which 50 % red blood cell lysis occurs. IC_{50} denotes the polymer concentration at which more than 50 % of 3T3 fibroblast cells remain viable, serving as a threshold for cytotoxicity. The hemolytic selectivity index (**SI_H**) is defined as the ratio of HC_{50} (μg mL^{-1}) to MIC_{90} (μg mL^{-1}), and the cytotoxicity selectivity index (**SI_C**) is calculated as the ratio of IC_{50} (μg mL^{-1}) to MIC_{90} (μg mL^{-1}).

2.8. Fungicidal potency of the polymers

To further evaluate the antifungal activity of these terpolymers and their end-group functionalized analogs, we determined their minimum fungicidal concentrations (MFCs) by back-plating following MIC determination. [14] While most terpolymers demonstrated potent inhibitory effects, their MFCs were consistently 4–8 times higher than their corresponding MICs across tested *Candida* species (the results for the *C. albicans* lab strain are shown in Table 4, and the results for the *C. albicans* clinical isolate, *C. tropicalis,* and *C. parapsilosis* are shown in **Supporting Information, Table S3**, indicating that the polymers kill *Candida* species in a concentration-dependent manner. This behavior is consistent with the clinical profile of the fungicidal drug amphotericin B and aligns with previously reported data. [14,38] In contrast, fluconazole and caspofungin exhibited fungistatic activity under our acidic assay conditions (**Supporting Information, Table S3**), consistent with prior observations. [34,38].

2.9. Hemolytic activity

We next performed a red blood cell hemolysis assay to evaluate the hemocompatibility of the terpolymers expressed concentration required to lyse 50 % (HC_{50}; **Supporting Information**). [64] As shown in Table 4, most terpolymers exhibited excellent hemocompatibility, with HC_{50} values exceeding 2000 μg/mL. A notable exception was the short polymer *X*-B25 ($X_n \approx 10$), which induced moderate hemolysis at a concentration of 1 mg mL^{-1} (HC_{50} = 1000 μg mL^{-1}). We then examined

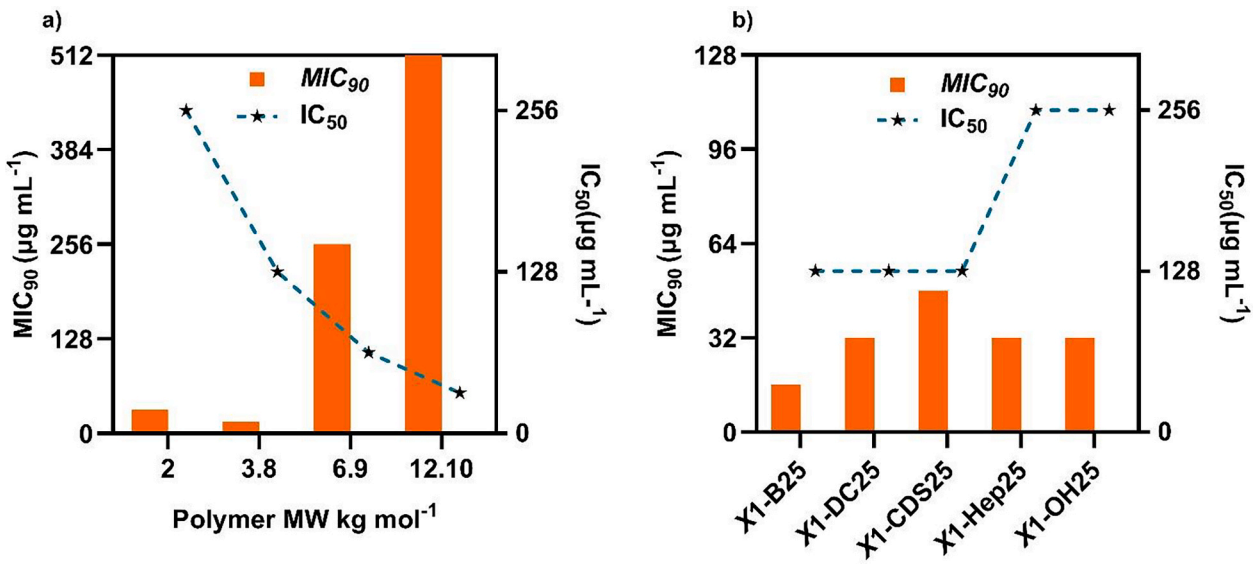


Fig. 4. Selectivity window of terpolymers shows the relationship between minimum inhibitory concentration (MIC_{90}) against *C. albicans* (SC5314) and cytotoxicity (IC_{50}) against mammalian 3 T3 cells. a) Effect of molecular weight polymers on therapeutic window. b) Selectivity windows of polymers with various end groups. ***Note:*** MIC_{90} represents the minimum concentration required to inhibit at least 90 % growth *ca1* = *C. albicans* (SC5314), and IC_{50} denotes the polymer concentration at which more than 50 % of 3T3 fibroblast cells remain viable, serving as a threshold for cytotoxicity.

the effect of the RAFT end group at varying hydrophobic content. Terpolymer synthesized with the three distinct RAFT agents showed no measurable hemolytic activity ($HC_{50} > 2000$ µg mL^{-1}) when formulated with 25 % of hydrophobic monomer content (*X*1-B25, *X*1-DC25, and *X*1-CDS25). However, at 30 % hydrophobic content, differences emerged depending on the end group. The polymer bearing cyanovaleric acid α-end group (*X*1-CDS30) displayed elevated hemolysis ($HC_{50} = 500$ µg mL^{-1}), whereas the corresponding polymer with a carboxylic acid α-end group (*X*1-DC30) retained excellent hemocompatibility ($HC_{50} > 2000$ µg mL^{-1}). This suggests that the introduction of a cyanovaleric acid *α*-end-group may enhance RBC interaction and contribute to increased hemolytic activity, a trend also observed for polymethacrylate polymers. [37].

Further differences were observed when comparing a shorter chain (C_4) and a longer chain (C_{12}) *ω*–end-group: the C_4-terminated polymers (*X*1-B30) showed increasing hemolytic activity ($HC_{50} = 1000$ µg mL^{-1}) compared with the C_{12}-terminated polymers (*X*1-DC30) under the same assay conditions. Similar studies on methacrylate copolymers and nylon-3 copolymers have reported a positive correlation between increasing *ω*–end-group hydrophobicity and elevated hemolysis. [44,65] Our findings expand upon this, revealing a more nuanced relationship when polymer hydrophobicity changes. Specifically, greater end-group hydrophobicity did not consistently correspond to increasing hemolysis (Table 4; **Supporting Information, Table S4**). Instead, hemocompatibility appears to be governed by the interplay between hydrophobic monomer content and the ClogP of end-groups (Table 1). Taken together, these results indicate that hemocompatibility is not solely determined by polymer hydrophobicity but is also influenced by both *α*–and *ω*–end-group design.

2.10. Cytotoxicity

The biocompatibility of the synthesized polymers was further assessed by evaluating mammalian cell viability and cytotoxicity using the MTT assay on 3T3 fibroblasts **(See Supporting Information)**. [66] The results are reported as an IC_{50} value, defined as the polymer concentration at which more than 50 % of 3T3 fibroblast cells remain viable, providing a threshold for cytotoxicity. Increasing the polymer molecular weight increased cytotoxicity (Table 4, **Supporting Information, Fig. S19**), consistent with previous findings on polyacrylamide polymer tested against a mouse embryonic fibroblast cell line. [14] *X*1-B25 (25 % hydrophobic monomer; $IC_{50} = 128$ µg mL^{-1}, Table 4), selected as a reference polymer for cytotoxicity comparisons due to its use in *ω*–end-group modification, showing an IC_{50} consistent with earlier reports. [34,38] Hydrophobic monomer content had a strong influence on cytotoxicity: reducing hydrophobic monomer content to 20 % decreased toxicity, while increasing it to 30 % markedly enhanced cytotoxicity (Table 4**)**. End-group differences further impacted cytotoxicity. Variation in the *α*–end-group carboxylic acid (–COOH) and cyanovaleric acid had minimal effect. However, *ω*–end-group substitution with hydrophilic hydroxyl (–OH) terminal or hydrophobic HepAm group led to a reduction in cytotoxicity (Table 4, **Supporting Information, Fig. S19**). Notably, replacing *ω*–end-group with a hydroxyl (–OH) in a polymer with 30 % hydrophobic content resulted in a fourfold reduction in cytotoxicity ($IC_{50} = 64$ µg mL^{-1} for *X*1-B30 vs. $IC_{50} = 256$ µg mL^{-1} for *X*1-OH30), highlighting the critical role of *ω*–end-group design in enhancing mammalian cell compatibility.

To integrate antifungal potency and mammalian cell compatibility into a single metric, the selectivity index (SI) was calculated as the ratio of IC_{50} to MIC (against *Candida* spp.) for each polymer **(Supporting Information, Table S5**). This metric provides a quantitative measure of the therapeutic window, with higher SI values reflecting greater fungal selectivity and improved safety margins. [28,67] SI values revealed that shorter polymers demonstrate improved cytotoxicity selectivity indices against *C. albicans* SC5314 (Fig. 4**)**: Consistent with recent work on PEGylated cationic ionenes, [63] lowering polymer molecular weight increased antifungal potency while reducing mammalian-cell toxicity. This reinforces that molecular weight is one of the principals determining factors of antifungal selectivity across diverse cationic polymer scaffolds. However, the shortest polymer *X*-B25 *(Xn ≈ 10)* showed notable hemolytic activity at a concentration of 1 mg mL^{-1}, limiting its biocompatibility (Table 3**)**. At 20 % hydrophobic monomer content, *X*-DC25 terpolymers bearing long C_{12} alkyl Z–end groups showed superior selectivity ($SI_C = 16$) compared to *X*-B20 polymer with short C_4 Z–end groups ($SI_C = 4$). Increasing the hydrophobic monomer content to 30 % reduced both hemocompatibility and 3T3 fibroblast cell selectivity. In addition, *ω*–end modification, including hydrophilic hydroxyl and hydrophobic heptyl acrylamide groups, confirmed that terminal group polarity also influences antifungal performance. Polymers with hydrophilic hydroxyl termini improved biocompatibility but reduced

antifungal potency, unless offset by higher backbone hydrophobicity (e.g., *X*1-B30; Table 3). Altogether, these results highlight that precise tuning of end group and polymer composition is essential for maximizing antifungal selectivity.

3. Conclusion

We synthesized a library of 16 well-defined ternary polyacrylamide terpolymers via one-pot PET-RAFT polymerization, employing three distinct RAFT agents combined with tailored ω–end-group modifications. Antifungal activity was inversely correlated with the molecular weight of the polymer against *C. parapsilosis*. Additionally, a polymer threshold effect was observed at around MW = 3.8 kg mol^{-1} (*X*1-B25, $X_n \approx 20$) against *C. albicans* and *C. tropicalis.* Shorter polymers demonstrate improved selectivity indices across tested *Candida* species; however, the lowest molecular weight analogue (*X*-B25, $X_n \approx 10$) showed reduced hemocompatibility. Polymer end-groups markedly influenced the biocompatibility profile. Terpolymer bearing cyanovaleric acid (CVA) α–end-groups displayed increased hemolytic activity, especially at higher hydrophobic monomer content, compared to their carboxylic acid counterparts. Moreover, the interplay between ω–end-group design and internal hydrophobic/hydrophilic balance emerged as a critical determinant of antifungal efficacy and biocompatibility. Collectively, this work demonstrates an integrated strategy to optimize the selectivity index of antifungal polymers through rational molecular design and end-group engineering. Future development of peptide-mimicking antifungal polymers will benefit from explicitly integrating end-group design into SAR models.

CRediT authorship contribution statement

Hatu Gmedhin: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation. **Peifeng Wu:** Writing – review & editing, Investigation, Formal analysis. **Zi Gu:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis. **Megan D. Lenardon:** Writing – review & editing, Supervision, Methodology, Formal analysis. **Cyrille Boyer:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

H.G. acknowledges the receipt of an Australian Government Research Training Program (RTP) Scholarship. C.B. is the recipient of an Australian Research Council (ARC) - Australian Laureate Fellowship (Project no. FL220100016) funded by the Australian Government. We thank Jonathan Yeow for his assistance in setting up the aqueous SEC measurement. The authors also express their gratitude to Eh Hau Pan, Camillo Taraborrelli, and the NMR facility within the Mark Wainwright Analytical Centre (MWAC) at UNSW for providing technical support and maintaining the necessary instruments.

Appendix A. Supplementary data

Figs. S1-S19, Tables S1-S5, and the experimental methods and materials sections are available in the supporting information. Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2025.114384.

Data availability

Data will be made available on request.

References

[1] A. Vitiello, F. Ferrara, M. Boccellino, A. Ponzo, C. Cimmino, E. Comberiati, A. Zovi, S. Clemente, M. Sabbatucci, Antifungal drug resistance: an emergent health threat, Biomedicines 11 (4) (2023) 1063.
[2] N. van Rhijn, S. Arikan-Akdagli, J. Beardsley, F. Bongomin, A. Chakrabarti, S. C. Chen, T. Chiller, A.L. Colombo, N.P. Govender, A. Alastruey-Izquierdo, Beyond bacteria: the growing threat of antifungal resistance, Lancet 404 (10457) (2024) 1017–1018.
[3] K.S. Ikuta, T. Meštrović, M. Naghavi, Global incidence and mortality of severe fungal disease, Lancet Infect. Dis. 24 (5) (2024) e268–e.
[4] P.B. Mazi, M.A. Olsen, D. Stwalley, A.M. Rauseo, C. Ayres, W.G. Powderly, A. Spec, Attributable Mortality of *Candida* Bloodstream Infections in the Modern Era: A Propensity Score Analysis, Clin. Infect. Dis. 75 (6) (2022) 1031–1036.
[5] C. Firacative, Invasive fungal disease in humans: are we aware of the real impact? Memórias do Instituto Oswaldo Cruz 115 (2020) e200430.
[6] A.M. Borman, A. Abdolrasouli, E.M. Johnson, Name changes for fungi of medical importance, 2022–2024, J. Clin. Microbiol. (2025) e02041–e2124.
[7] A.F. Makled, S.A.M. Ali, A.Z. Labeeb, S.S. Salman, D.Z.M. Shebl, S.G. Hegazy, M. S. Sabal, Characterization of *Candida* species isolated from clinical specimens: insights into virulence traits, antifungal resistance and molecular profiles, BMC Microbiol. 24 (1) (2024) 388.
[8] World Health Organization, in: Fungal priority pathogens list to guide research, development and public health action, 2022. https://www.who.int/publications/i/item/9789240060241.
[9] C. Lass-Flörl, S.S. Kanj, N.P. Govender, G.R. Thompson, L. Ostrosky- Zeichner, M. A. Govrins, Invasive candidiasis, Nat. Rev. Dis. Primers 10 (1) (2024) 20.
[10] J. Berman, D.J. Krysan, Drug resistance and tolerance in fungi, Nat. Rev. Microbiol. 18 (6) (2020) 319–331.
[11] P. Osset-Trénor, A. Pascual-Ahuir, M. Proft, Fungal Drug Response and Antimicrobial Resistance, Journal of Fungi 9 (5) (2023) 565.
[12] S. Wu, R. Song, T. Liu, C. Li, Antifungal therapy: Novel drug delivery strategies driven by new targets, Adv. Drug Deliv. Rev. 199 (2023) 114967.
[13] J.R. Perfect, The antifungal pipeline: a reality check, Nat. Rev. Drug Discov. 16 (9) (2017) 603–616.
[14] S. Schaefer, T.T.P. Pham, S. Brunke, B. Hube, K. Jung, M.D. Lenardon, C. Boyer, Rational Design of an Antifungal Polyacrylamide Library with Reduced Host-Cell Toxicity, ACS Appl. Mater. Interfaces 13 (23) (2021) 27430–27444.
[15] R. Liu, X. Chen, S.P. Falk, B.P. Mowery, A.J. Karlsson, B. Weisblum, S.P. Palecek, K. S. Masters, S.H. Gellman, Structure-Activity Relationships among Antifungal Nylon-3 Polymers: Identification of Materials Active against Drug-Resistant Strains of *Candida albicans*, J. Am. Chem. Soc. 136 (11) (2014) 4333–4342.
[16] S. Schaefer, N. Corrigan, S. Brunke, M.D. Lenardon, C. Boyer, Combatting Fungal Infections: Advances in Antifungal Polymeric Nanomaterials, Biomacromolecules 25 (9) (2024) 5670–5701.
[17] R. Liu, X. Chen, S.P. Falk, K.S. Masters, B. Weisblum, S.H. Gellman, Nylon-3 Polymers Active against Drug-Resistant *Candida albicans* Biofilms, J. Am. Chem. Soc. 137 (6) (2015) 2183–2186.
[18] W. Jiang, M. Zhou, S. Chen, J. Xie, M. Chen, H. Zhang, Y. Wu, X. Chen, R. Liu, Peptide-mimicking poly (2-oxazoline) s possessing potent antifungal activity and BBB penetrating property to treat invasive infections and meningitis, J. Am. Chem. Soc. 145 (47) (2023) 25753–25765.
[19] Y. Lee, C. Boyer, M.S. Kwon, Photocontrolled RAFT polymerization: past, present, and future, Chem. Soc. Rev. 52 (9) (2023) 3035–3097.
[20] G. Moad, E. Rizzardo, S.H. Thang, Living Radical Polymerization by the RAFT Process - A Second Update, Aust. J. Chem. 62 (11) (2009) 1402.
[21] C. Boyer, V. Bulmus, T.P. Davis, V. Ladmiral, J. Liu, S. Perrier, Bioapplications of RAFT Polymerization, Chem. Rev. 109 (11) (2009) 5402–5436.
[22] N. Corrigan, K. Jung, G. Moad, C.J. Hawker, K. Matyjaszewski, C. Boyer, Reversible-deactivation radical polymerization (Controlled/living radical polymerization): From discovery to materials design and applications, Prog. Polym. Sci. 111 (2020) 101311.
[23] S. Perrier, 50th Anniversary Perspective : RAFT Polymerization—A User Guide, Macromolecules 50 (19) (2017) 7433–7447.
[24] H. Willcock, R.K. O'Reilly, End group removal and modification of RAFT polymers, Polym. Chem. 1 (2) (2010) 149–157.
[25] J.Y.T. Chong, D.J. Keddie, A. Postma, X. Mulet, B.J. Boyd, C.J. Drummond, RAFT preparation and the aqueous self-assembly of amphiphilic poly(octadecyl acrylate)-block -poly(polyethylene glycol methyl ether acrylate) copolymers, Colloids Surf A Physicochem Eng Asp 470 (2015) 60–69.
[26] D. Wyers, T. Jirapanjawat, J.F. Quinn, M.R. Whittaker, C. Greening, T. Junkers, Narrowing down chain length effects on the antibacterial action of guanylated oligomers, Polym. Chem. 14 (17) (2023) 2126–2134.
[27] J. Roh, K.-T. Huang, A. Choudhury, T. Quan, C.-J. Huang, P. Kumar, C. Boyer, Machine learning-guided design of high-performance antifungal polymers against *Candida albicans*, ACS Polymers Au 5 (5) (2025) 557–569.
[28] S. Schaefer, R. Vij, J.L. Sprague, S. Austermeier, H. Dinh, P.R. Judzewitsch, S. Müller-Loennies, T. Lopes Silva, E. Seemann, B. Qualmann, A synthetic peptide mimic kills *Candida albicans* and synergistically prevents infection, Nat. Commun. 15 (1) (2024) 6818.

[29] J. Pachla, R.J. Kopiasz, G. Marek, W. Tomaszewski, A. Głogowska, K. Drężek, S. Kowalczyk, R. Podgórski, B. Butruk-Raszeja, T. Ciach, J. Mierzejewska, A. Plichta, E. Augustynowicz-Kopeć, D. Jańczewski, Polytrimethylenimines: Highly Potent Antibacterial Agents with Activity and Toxicity Modulated by the Polymer Molecular Weight, Biomacromolecules 24 (5) (2023) 2237–2249.

[30] Y. Yang, Z. Cai, Z. Huang, X. Tang, X. Zhang, Antimicrobial cationic polymers: from structural design to functional control, Polym. J. 50 (1) (2018) 33–44.

[31] Y. Xue, H. Xiao, Y. Zhang, Antimicrobial polymeric materials with quaternary ammonium and phosphonium salts, Int. J. Mol. Sci. 16 (2) (2015) 3626–3655.

[32] F. Padilla-Garfias, L. Ríos-Cifuentes, N.S. Sánchez, M. Calahorra, A. Peña, Study of the mechanism of ε-poly-l-lysine as an antifungal on *Candida albicans* and *Saccharomyces cerevisiae*, Biochim. Biophys. Acta Gen. Subj. 1866 (10) (2022) 130197.

[33] P. Pham, S. Oliver, C. Boyer, Design of Antimicrobial Polymers, Macromol. Chem. Phys. 224 (3) (2023) 2200226.

[34] S. Schaefer, D. Melodia, N. Corrigan, M.D. Lenardon, C. Boyer, Effect of Star Topology Versus Linear Polymers on Antifungal Activity and Mammalian Cell Toxicity, Macromol. Biosci. 24 (5) (2024) e2300452.

[35] C. Gonnot, M.B. Hassan Mahboob, M. Aumond, J.R. Tait, K. Nay, K. Nazemi, H. Floyd, J. Zuegg, F. Boeda, C.B. Landersdorfer, J.F. Quinn, L. Fontaine, M. R. Whittaker, V. Montembault, Cyclic and linear cationic polymers based on metathesis polymerization for antibacterial and antifungal Applications, Eur. Polym. J. 230 (2025) 113875.

[36] M. Scherger, Y.A. Pilger, J. Stickdorn, P. Komforth, S. Schmitt, K. Koynov, H. J. Räder, L. Nuhn, Efficient self-immolative RAFT end group modification for macromolecular immunodrug delivery, Biomacromolecules 24 (5) (2023) 2380–2391.

[37] T.D. Michl, K.E.S. Locock, N.E. Stevens, J.D. Hayball, K. Vasilev, A. Postma, Y. Qu, A. Traven, M. Haeussler, L. Meagher, H.J. Griesser, RAFT-derived antimicrobial polymethacrylates: elucidating the impact of end-groups on activity and cytotoxicity, Polym. Chem. 5 (19) (2014) 5813–5822.

[38] H. Gmedhin, S. Schaefer, N. Corrigan, P. Wu, Z. Gu, M.D. Lenardon, C. Boyer, Effect of Defined Block Sequence Terpolymers on Antifungal Activity and Biocompatibility, Macromol. Biosci. 25 (4) (2025) e2400429.

[39] S. Xu, P. Tan, Q. Tang, T. Wang, Y. Ding, H. Fu, Y. Zhang, C. Zhou, M. Song, Q. Tang, Enhancing the stability of antimicrobial peptides: From design strategies to applications, Chem. Eng. J. 475 (2023) 145923.

[40] M. Shahmiri, A. Mechler, The role of C-terminal amidation in the mechanism of action of the antimicrobial peptide aurein 1.2, The EuroBiotech Journal 4 (1) (2020) 25–31.

[41] J. Song, S. Zhang, J. Xing, L. Zhang, J. Wang, A. Shan, Optimizing therapeutic efficacy of antifungal peptides via strategic terminal amino acid modification, J. Adv. Res. 74 (2025) 555–570.

[42] R. Nikookar Golestani, E. Ghods, M. Rostamian, H. Madanchi, A.F. Talebi, Investigating the antimicrobial activity, cytotoxicity, and action mechanism of acylated and amidated derivatives of AurH1 antifungal peptide, BMC Microbiol. 23 (1) (2023) 332.

[43] J. Zhang, M.J. Markiewicz, B.P. Mowery, B. Weisblum, S.S. Stahl, S.H. Gellman, C-terminal functionalization of nylon-3 polymers: effects of C-terminal groups on antibacterial and hemolytic activities, Biomacromolecules 13 (2) (2012) 323–331.

[44] E.T. Nadres, H. Takahashi, K. Kuroda, Radical-medicated end-group transformation of amphiphilic methacrylate random copolymers for modulation of antimicrobial and hemolytic activities, J. Polym. Sci. A Polym. Chem. 55 (2) (2017) 304–312.

[45] G. Opiyo, J. Jin, Recent progress in switchable RAFT agents: Design, synthesis and application, Eur. Polym. J. 159 (2021) 110713.

[46] S. Schaefer, D. Melodia, C. Pracey, N. Corrigan, M.D. Lenardon, C. Boyer, Mimicking Charged Host-Defense Peptides to Tune the Antifungal Activity and Biocompatibility of Amphiphilic Polymers, Biomacromolecules 25 (2) (2024) 871–889.

[47] P. Pham, S. Oliver, E.H.H. Wong, C. Boyer, Effect of hydrophilic groups on the bioactivity of antimicrobial polymers, Polym. Chem. 12 (39) (2021) 5689–5703.

[48] C.P. Jesson, C.M. Pearce, H. Simon, A. Werner, V.J. Cunningham, J.R. Lovett, M. J. Smallridge, N.J. Warren, S.P. Armes, H 2 O 2 Enables Convenient Removal of RAFT End-Groups from Block Copolymer Nano-Objects Prepared via Polymerization-Induced Self-Assembly in Water, Macromolecules 50 (1) (2017) 182–191.

[49] J.M. Spruell, B.A. Levy, A. Sutherland, W.R. Dichtel, J.Y. Cheng, J.F. Stoddart, A. Nelson, Facile postpolymerization end-modification of RAFT polymers, J. Polym. Sci. A Polym. Chem. 47 (2) (2009) 346–356.

[50] P.J. Roth, C. Boyer, A.B. Lowe, T.P. Davis, RAFT Polymerization and Thiol Chemistry: A Complementary Pairing for Implementing Modern Macromolecular Design, Macromol. Rapid Commun. 32 (15) (2011) 1123–1143.

[51] G. Moad, Y.K. Chong, A. Postma, E. Rizzardo, S.H. Thang, Advances in RAFT polymerization: the synthesis of polymers with defined end-groups, Polymer 46 (19) (2005) 8458–8468.

[52] C. Struyfs, B.P.A. Cammue, K. Thevissen, Membrane-Interacting Antifungal Peptides, Front. Cell Dev. Biol. 9 (2021) 649875.

[53] S. Parambath, A. Dao, H.Y. Kim, S. Zawahir, A. Alastruey Izquierdo, E. Tacconelli, N. Govender, R. Oladele, A. Colombo, T. Sorrell, P. Ramon-Pardo, T. Fusire, V. Gigante, H. Sati, C.O. Morrissey, J.-W. Alffenaar, J. Beardsley, *Candida albicans* —A systematic review to inform the World Health Organization fungal priority pathogens list, Med. Mycol. 62 (6) (2024) myae045.

[54] C. Keighley, H.Y. Kim, S. Kidd, S.C.A. Chen, A. Alastruey, A. Dao, F. Bongomin, T. Chiller, R. Wahyuningsih, A. Forastiero, A. Al-Nuseirat, P. Beyer, V. Gigante, J. Beardsley, H. Sati, C.O. Morrissey, J.-W. Alffenaar, *Candida tropicalis* —A systematic review to inform the World Health Organization of a fungal priority pathogens list, Med. Mycol. 62 (6) (2024) myae040.

[55] M. Asogan, H.Y. Kim, S. Kidd, A. Alastruey-Izquierdo, N.P. Govender, A. Dao, J.-H. Shin, J. Heim, N.P. Ford, V. Gigante, H. Sati, C.O. Morrissey, J.-W. Alffenaar, J. Beardsley, *Candida parapsilosis* : A systematic review to inform the World Health Organization fungal priority pathogens list, Med. Mycol. 62 (6) (2024) myad131.

[56] F.M. Anderson, N.D. Visser, K.R. Amses, A. Hodgins-Davis, A.M. Weber, K. M. Metzner, M.J. McFadden, R.E. Mills, M.J. O'Meara, T.Y. James, T.R. O'Meara, *Candida albicans* selection for human commensalism results in substantial within-host diversity without decreasing fitness for invasive disease, PLoS Biol. 21 (5) (2023) e3001822.

[57] A.M. Gillum, E.Y.H. Tsay, D.R. Kirsch, Isolation of the *Candida albicans* gene for orotidine-5′-phosphate decarboxylase by complementation of *S. cerevisiae ura3* and *E. coli pyrF* mutations, Mol. Gen. Genet. 198 (1) (1984) 179–182.

[58] D.M. MacCallum, L. Castillo, K. Nather, C.A. Munro, A.J.P. Brown, N.A.R. Gow, F. C. Odds, Property Differences among the Four Major *Candida albicans* Strain Clades, Eukaryot. Cell 8 (3) (2009) 373–387.

[59] F.C. Odds, M.F. Hanson, A.D. Davidson, M.D. Jacobsen, P. Wright, J.A. Whyte, N.A. R. Gow, B.L. Jones, One year prospective survey of *Candida* bloodstream infections in Scotland, J. Med. Microbiol. 56 (8) (2007) 1066–1075.

[60] D. Briand, E. Dubreucq, P. Galzy, Functioning and regioselectivity of the lipase of *Candida parapsilosis* (Ashford) Langeron and Talice in aqueous medium. New interpretation of regioselectivity taking acyl migration into account, Eur. J. Biochem. 228 (1) (1995) 169–175.

[61] CLSI. Reference method for broth dilution antifungal susceptibility testing of yeasts, 4th ed., Clinical and Laboratory Standards Institute, Wayne, PA, 2017.

[62] L.G.S. Garcia, G.M.d.M. Guedes, M.L.Q. da Silva, D.S.C.M. Castelo-Branco, J.J. C. Sidrim, R.d.A. Cordeiro, M.F.G. Rocha, R.S. Vieira, R.S.N. Brilhante, Effect of the molecular weight of chitosan on its antifungal activity against *Candida* spp. in planktonic cells and biofilm, Carbohydr. Polym. 195 (2018) 662–669.

[63] J.M. Kurzyna, R.J. Kopiasz, M. Paul, M. Flont, P. Baranowska, J. Mierzejewska, K. Drężek, W. Tomaszewski, E. Jastrzębska, D. Jańczewski, Unlocking the Potential: PEGylation and Molecular Weight Reduction of Ionenes for Enhanced Antifungal Activity and Biocompatibility, Macromol. Biosci. 24 (9) (2024) 2400032.

[64] P.R. Judzewitsch, L. Zhao, E.H.H. Wong, C. Boyer, High-Throughput Synthesis of Antimicrobial Copolymers and Rapid Evaluation of Their Bioactivity, Macromolecules 52 (11) (2019) 3975–3986.

[65] B.P. Mowery, A.H. Lindner, B. Weisblum, S.S. Stahl, S.H. Gellman, Structure−activity Relationships among Random Nylon-3 Copolymers That Mimic Antibacterial Host-Defense Peptides, J. Am. Chem. Soc. 131 (28) (2009) 9735–9745.

[66] B. Li, G. Hao, B. Sun, Z. Gu, Z.P. Xu, Engineering a Therapy-Induced "Immunogenic Cancer Cell Death" Amplifier to Boost Systemic Tumor Elimination, Adv. Funct. Mater. 30 (22) (2020) 1909745.

[67] R. Liu, X. Chen, Z. Hayouka, S. Chakraborty, S.P. Falk, B. Weisblum, K.S. Masters, S.H. Gellman, Nylon-3 Polymers with Selective Antifungal Activity, J. Am. Chem. Soc. 135 (14) (2013) 5270–5273.